Randomized and low-rank approximation

Random column subsets of arbitrary fixed-sparsity matrices

Difficulty: challenging **Importance:** interesting to the community

Status: open conjecture in a preprint updated 2026-07-08.

Problem statement

Fix an integer $s \geq 2$ and $C \geq 1$. Suppose $r_k \leq k$, $k/r_k \to C$, and $n_k/r_k \to \infty$. For every deterministic sequence $M_k \in \{-1, 0, 1\}^{k \times n_k}$ with exactly s nonzero entries per column, and a uniformly random r_k-element column set I_k, prove or disprove

$$\forall \eta > 0, \qquad \Pr\{\sigma_{\min}((M_k)_{:I_k}) > \eta\} \longrightarrow 0.$$

Here $\sigma_{\min}(B) = \inf_{\|x\|_2=1} \|Bx\|_2$. There is no hypothesis controlling intersections of column supports.

Reference

Han Huang, Mark Rudelson, and Konstantin Tikhomirov, *Well-Invertible Column Subsets of Sparse Matrices Are Rare*, §7, Conjecture 7.1; compare Theorem 1.3 for the additional structural hypothesis in the proved result.

Status check

Searches included *"unrestricted deterministic sparse sketches" conjecture* and the exact paper title with *2026*. No subsequent resolution was located. The paper's disproof of a particular SparseStack conjecture does not establish its own unrestricted deterministic conjecture.